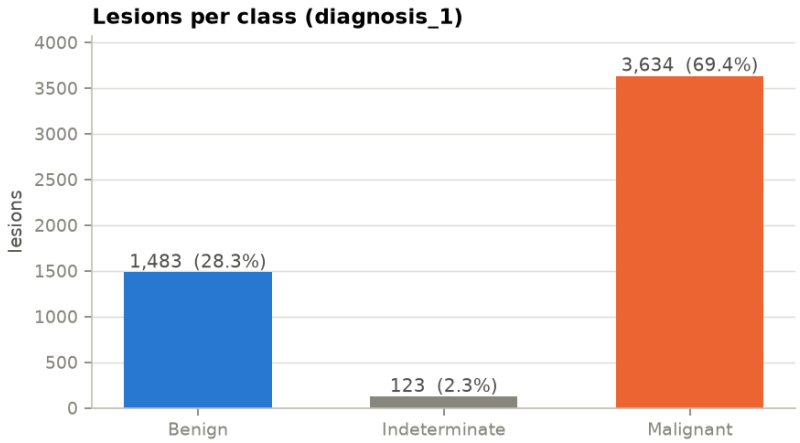
MILK10k — EDA first outputs

Computer Vision & Speech Recognition · Joshua Cole · code: https://github.com/jcole-jpg/Computer-Vision-and-Speech-Recognition

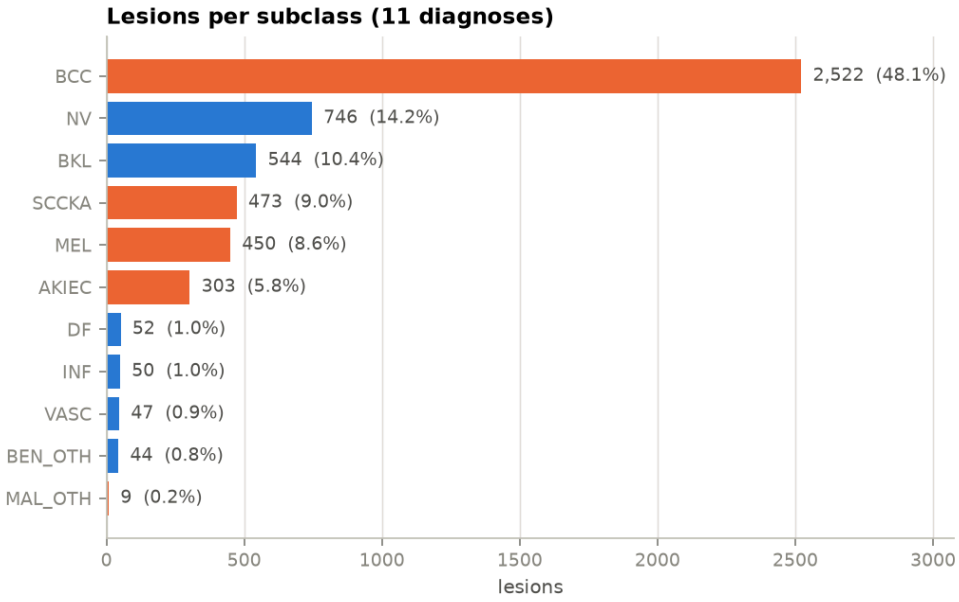
Unit = lesion (one row per lesion, 5,240 lesions = 10,480 images: one clinical close-up and one dermatoscopic image each). Age and sex belong to the lesion, so image counts are exactly twice the lesion counts. Source: `data/processed/lesions.csv`, built from `metadata.csv` + `training_gt.csv`; script `scripts/build_eda_first_outputs.py`.

1. Samples per class (diagnosis_1)



class (diagnosis_1)	lesions	share	images
Benign	1,483	28.3%	2,966
Indeterminate	123	2.3%	246
Malignant	3,634	69.4%	7,268

2. Samples per subclass (11 diagnoses)

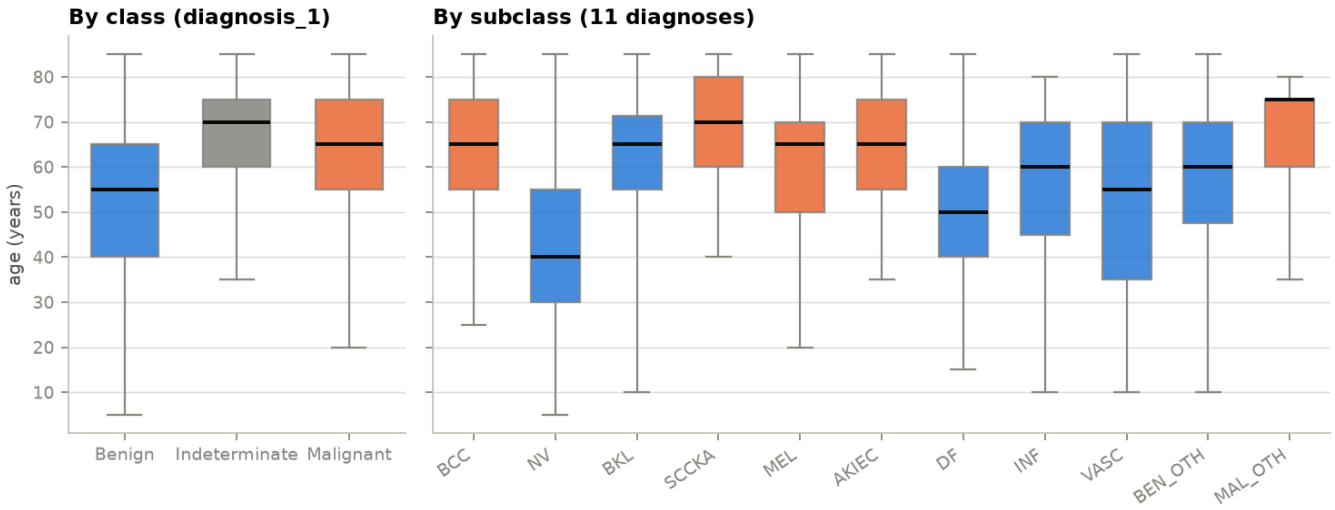


subclass	name	diagnosis_1	lesions	share
BCC	Basal cell carcinoma	Malignant	2,522	48.1%
NV	Melanocytic nevus	Benign	746	14.2%
BKL	Benign keratinocytic lesion	Benign	544	10.4%
SCCKA	Squamous cell carcinoma / keratoacanthoma	Malignant	473	9.0%
MEL	Melanoma	Malignant	450	8.6%
AKIEC	Actinic keratosis / intraepidermal carcinoma	Malignant (180) / Indeterminate (123)	303	5.8%
DF	Dermatofibroma	Benign	52	1.0%
INF	Inflammatory / infectious	Benign	50	1.0%
VASC	Vascular lesion / hemorrhage	Benign	47	0.9%
BEN_OTH	Other benign proliferations	Benign	44	0.8%
MAL_OTH	Other malignant proliferations	Malignant	9	0.2%

Malignant is 69% of the lesions and Benign 28%; BCC alone is 48%. The rarest subclass, MAL_OTH, has 9 lesions (280× fewer than BCC). This is a referral population, not a screening one. AKIEC is split between two classes: the 123 Indeterminate lesions are all AKIEC.

3. Age range per class

Age range per class — box = middle 50 %, line = median, whiskers = youngest / oldest



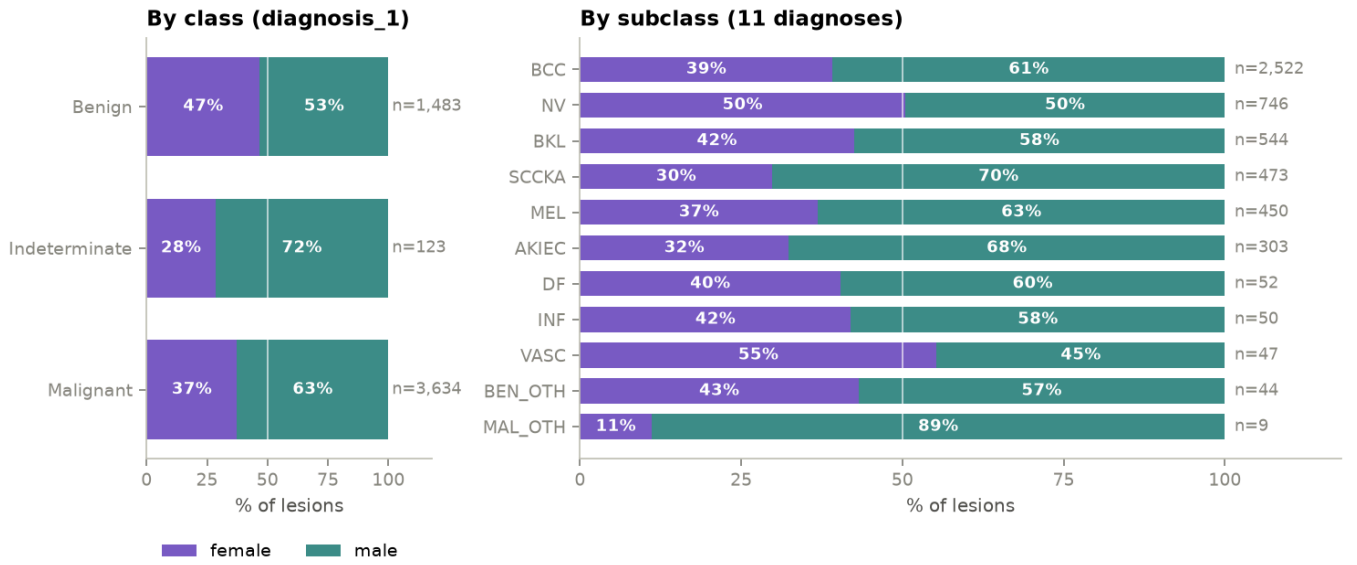
class (diagnosis_1)	n	missing	min	Q1	median	Q3	max	mean ± sd
Benign	1,483	20	5	40	55	65	85	51.6 ± 18.8
Indeterminate	123	0	35	60	70	75	85	67.8 ± 10.9
Malignant	3,634	0	20	55	65	75	85	65.1 ± 12.4

subclass	n	missing	min	Q1	median	Q3	max	mean ± sd
BCC	2,522	0	25	55	65	75	85	64.9 ± 11.8
NV	746	17	5	30	40	55	85	42.0 ± 17.7
BKL	544	0	10	55	65	71	85	63.7 ± 12.4
SCCKA	473	0	40	60	70	80	85	69.5 ± 11.2
MEL	450	0	20	50	65	70	85	61.8 ± 14.8
AKIEC	303	0	35	55	65	75	85	65.9 ± 12.3
DF	52	0	15	40	50	60	85	50.4 ± 14.4
INF	50	0	10	45	60	70	80	55.6 ± 17.3
VASC	47	2	10	35	55	70	85	52.0 ± 20.5
BEN_OTH	44	1	10	48	60	70	85	56.6 ± 19.4
MAL_OTH	9	0	35	60	75	75	80	66.7 ± 13.9

20 lesions have no recorded age (NV: 17, VASC: 2, BEN_OTH: 1); they are left out of the statistics but counted in *n*.

Median age runs from 40 (NV) to 75 (MAL_OTH). Benign lesions are younger (median 55) than Malignant (65) and Indeterminate (70); ages span 5–85 overall.

4. Sex distribution per class



class (diagnosis_1)	female	male	% female	% male
Benign	694	789	46.8	53.2
Indeterminate	35	88	28.5	71.5
Malignant	1,359	2,275	37.4	62.6

subclass	female	male	% female	% male
BCC	988	1,534	39.2	60.8
NV	376	370	50.4	49.6
BKL	231	313	42.5	57.5
SCCKA	141	332	29.8	70.2
MEL	166	284	36.9	63.1
AKIEC	98	205	32.3	67.7
DF	21	31	40.4	59.6
INF	21	29	42.0	58.0
VASC	26	21	55.3	44.7
BEN_OTH	19	25	43.2	56.8
MAL_OTH	1	8	11.1	88.9

Overall 40% of lesions are female. The most male-skewed subclass is MAL_OTH (11% female) and the most female-skewed is VASC (55% female); Indeterminate lesions are 71.5% male.